

GenomicRanges - seqinfo

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Dependencies

This document has the following dependencies:

```
library(GenomeInfoDb)
library(GenomicRanges)
```

Use the following commands to install these packages in R.

```
source("http://www.bioconductor.org/biocLite.R")
biocLite(c("GenomeInfoDb", "GenomicRanges"))
```

Corrections

Improvements and corrections to this document can be submitted on its [GitHub](#) in its [repository](#).

Overview

The GRanges class contains seqinfo information about the length and the names of the chromosomes. Here we will briefly discuss strategies for harmonizing this information.

The *GenomeInfoDb* package addresses a seemingly small, but consistent problem: different online resources use different naming conventions for chromosomes. In more technical jargon, this package helps keeping your `seqinfo` and `seqlevels` harmonized.

Other Resources

- The vignette from the [GenomeInfoDb webpage](#).

Drop and keep seqlevels

It is common to want to remove `seqlevels` from a `GRanges` object. Here are some equivalent methods

```
gr <- GRanges(seqnames = c("chr1", "chr2"),
              ranges = IRanges(start = 1:2, end = 4:5))
seqlevels(gr, force=TRUE) <- "chr1"
gr
```

```
## GRanges object with 1 range and 0 metadata columns:
##      seqnames      ranges strand
##      <Rle> <IRanges> <Rle>
## [1]    chr1    [1, 4]      *
## -----
## seqinfo: 1 sequence from an unspecified genome; no seqlengths
```

In *GenomeInfoDb* (loaded when you load *GenomicRanges*) you find `dropSeqlevels()` and `keepSeqlevels()`.

```
gr <- GRanges(seqnames = c("chr1", "chr2"),
              ranges = IRanges(start = 1:2, end = 4:5))
dropSeqlevels(gr, "chr1")
```

```
## GRanges object with 1 range and 0 metadata columns:
##      seqnames      ranges strand
##      <Rle> <IRanges> <Rle>
## [1]    chr2    [2, 5]      *
## -----
## seqinfo: 1 sequence from an unspecified genome; no seqlengths
```

```
keepSeqlevels(gr, "chr2")
```

```
## GRanges object with 1 range and 0 metadata columns:
##      seqnames      ranges strand
##      <Rle> <IRanges> <Rle>
## [1]      chr2      [2, 5]      *
## -----
## seqinfo: 1 sequence from an unspecified genome; no seqlengths
```

You can also just get rid of weird looking chromosome names with `keepStandardChromosomes()`.

```
gr <- GRanges(seqnames = c("chr1", "chrU345"),
              ranges = IRanges(start = 1:2, end = 4:5))
keepStandardChromosomes(gr)
```

```
## GRanges object with 1 range and 0 metadata columns:
##      seqnames      ranges strand
##      <Rle> <IRanges> <Rle>
## [1]      chr1      [1, 4]      *
## -----
## seqinfo: 1 sequence from an unspecified genome; no seqlengths
```

Changing style

It is an inconvenient truth that different online resources uses different naming convention for chromosomes. This can even be different from organism to organism. For example, for the fruitfly (*Drosophila Melanogaster*) NCBI and Ensembl uses “2L” and UCSC uses “chr2L”. But NCBI and Ensembl differs on some contigs: NCBI uses “Un” and Ensembl used “U”.

```
gr <- GRanges(seqnames = "chr1", ranges = IRanges(start = 1:2, width = 2
```



Let us remap

```
newStyle <- mapSeqlevels(seqlevels(gr), "NCBI")
gr <- renameSeqlevels(gr, newStyle)
```

This can in principle go wrong, if the original set of seqlevels are inconsistent (not a single style).

The *GenomeInfoDb* also contains information for dropping / keeping various classes of chromosomes:

Using information from BSgenome packages

BSgenome packages contains seqinfo on their genome objects. This contains seqlengths and other information. An easy trick is to use these packages to correct your seqinfo information.

Hopefully, in Bioconductor 3.2 we will get support for seqlengths in *GenomeInfoDb* so we can avoid using the big genome packages.

SessionInfo

```
## R version 3.2.1 (2015-06-18)
## Platform: x86_64-apple-darwin13.4.0 (64-bit)
## Running under: OS X 10.10.5 (Yosemite)
##
## locale:
## [1] en_US.UTF-8/en_US.UTF-8/en_US.UTF-8/C/en_US.UTF-8/en_US.UTF-8
##
## attached base packages:
## [1] parallel stats4      methods      stats        graphics  grDevices utils
## [8] datasets  base
##
## other attached packages:
## [1] GenomicRanges_1.20.5 GenomeInfoDb_1.4.2  IRanges_2.2.7
## [4] S4Vectors_0.6.3      BiocGenerics_0.14.0 BiocStyle_1.6.0
## [7] rmarkdown_0.7
##
## loaded via a namespace (and not attached):
## [1] digest_0.6.8      formatR_1.2        magrittr_1.5       evaluate_0.7.2
## [5] stringi_0.5-5     xVector_0.8.0      tools_3.2.1        stringr_1.0.0
## [9] yaml_2.1.13       htmltools_0.2.6    knitr_1.11
```